User Manual for Spectrum Analysis Web Application

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Website: https://roshanchenglei.github.io/SA

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1. Introduction

The Spectrum Analysis Web Application is an interactive online platform designed to analyze spectral data of active galaxies. It performs continuum fitting, emission and absorption line analysis, and AGN-based modeling to estimate black hole mass and the size of the broad-line region (BLR). The tool runs entirely in a browser with no installation required.

Supported Browsers: Chrome, Edge, Firefox.

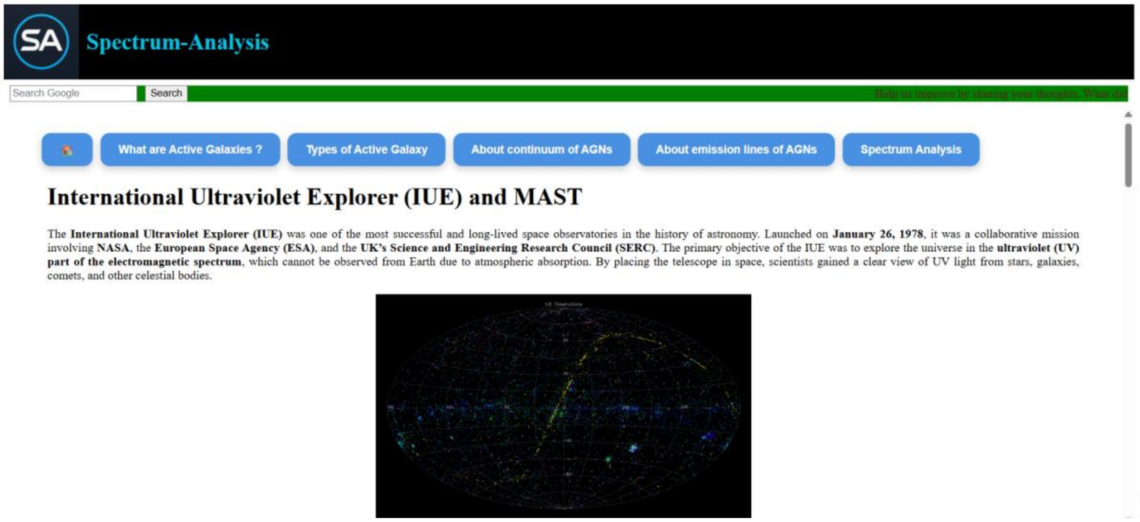
2. Getting Started

1. Open your preferred browser (Chrome, Edge, or Firefox).

2. Navigate to: https://roshanchenglei.github.io/SA

3. The homepage will appear as shown below.

Figure 1:



3. Uploading Input Data

The tool accepts plain text (.txt) files containing two columns: wavelength (in Ångström or nm) and flux (arbitrary units). Ensure your file is formatted correctly before uploading.

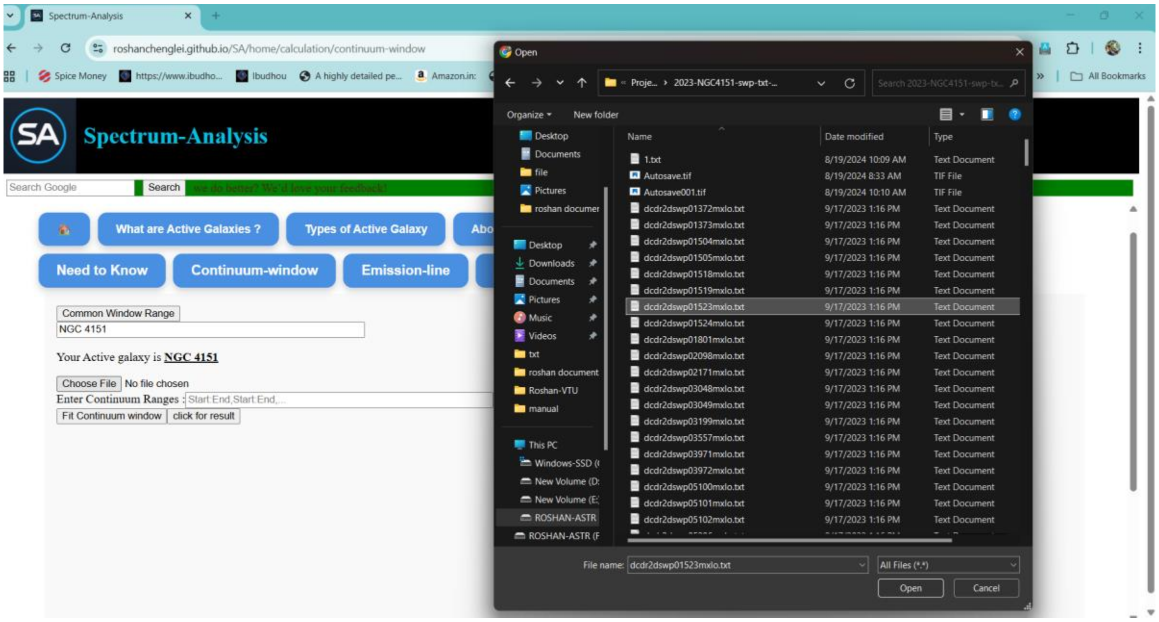
Example Input Format:

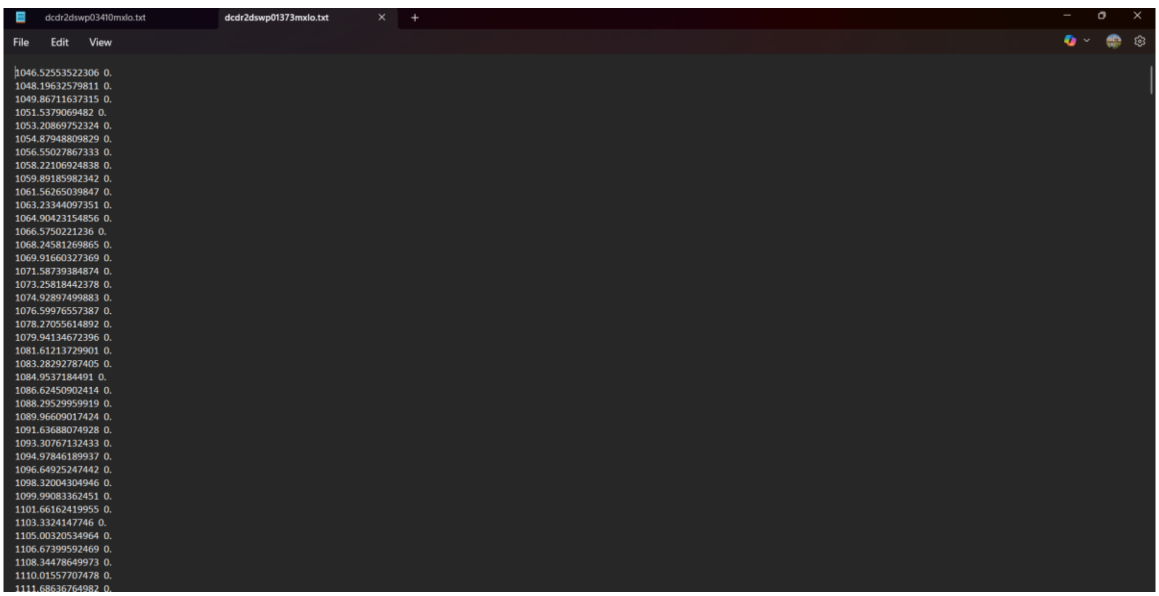
Wavelength (Å)	Flux
1300	2.45e-14
1310	2.50e-14

Steps to Upload:

- Click 'Choose File'.
- Select your .txt spectral data file.
- Enter the continuum ranges (e.g., 1200:1250, 1300:1350).
- Click 'Fit Continuum' to check fitting or 'Click for Result' to process.

Figure 2:





4. Continuum Analysis

This feature fits a power-law continuum to the spectral data within user-defined wavelength ranges. Users can visualize the fitted continuum curve and inspect residuals.

Steps:

- Click on 'Continuum-window'.
- Enter continuum window ranges (e.g., 1300:1350 , 1800:1850,...).
- Click 'Fit Continuum' to perform the fit.

Figure 3:

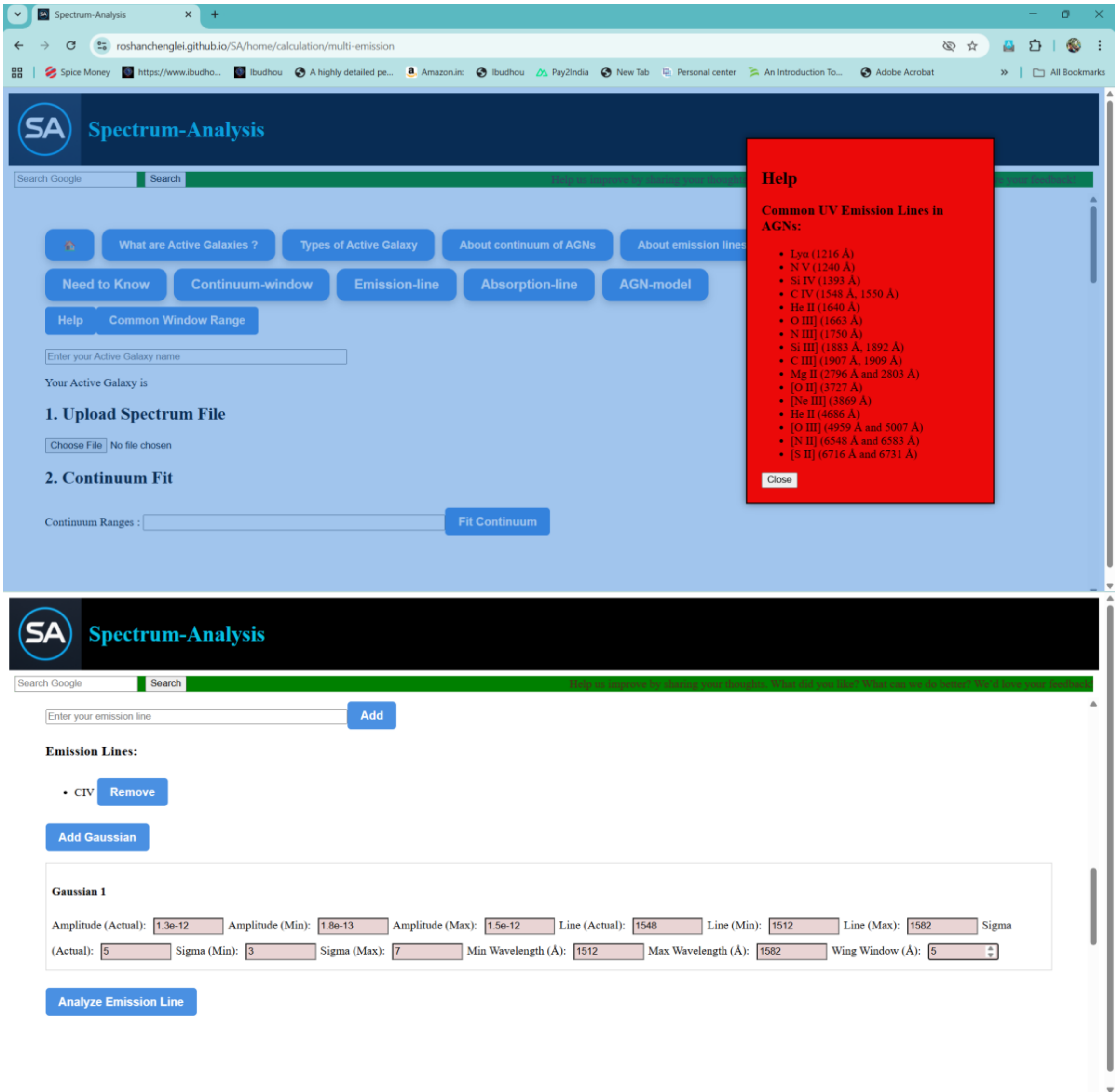
5. Emission-Line Analysis

This module identifies and models emission features in the spectrum using Gaussian fitting. Users can input multiple emission lines and view parameters such as central wavelength, flux, and FWHM.

Steps:

- Click on 'Emission-line'.
- Enter the emission line name and wavelength.
- If multiple lines, click 'Add Gaussian' for each additional component.
- Click 'Analyze Emission Line' to process.

Figure 4:

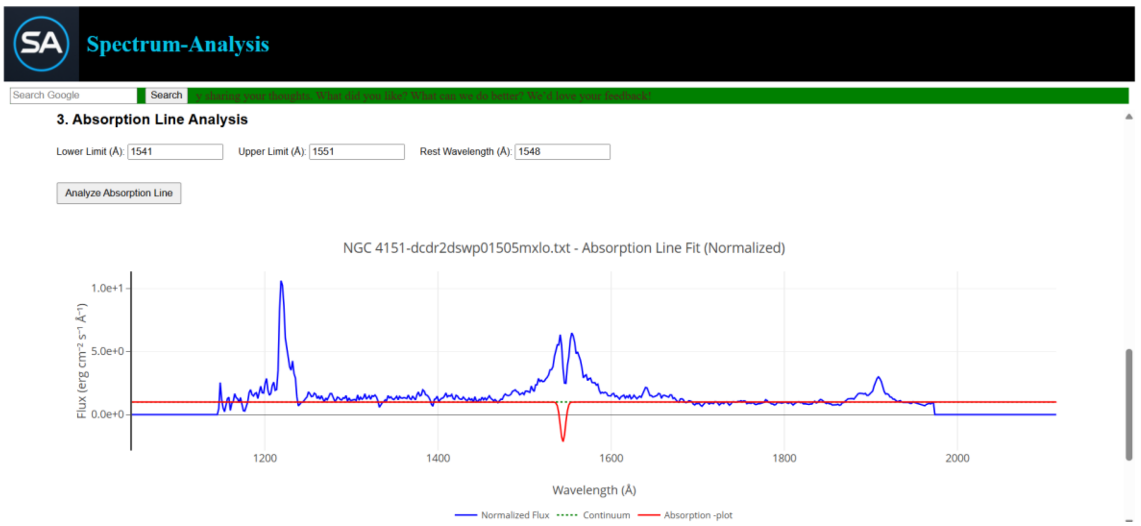


6. Absorption-Line Analysis

The absorption-line module allows users to identify absorption features and estimate equivalent widths or line depths.

Steps:

- Click on 'Absorption-line'.
- Select or enter the expected absorption region.
- Click 'Analyze Absorption Line' to process the data.



7. AGN Model: Black Hole Mass and BLR Size

This feature calculates the black hole mass (M_BH) and BLR radius (R_BLR) using emission-line parameters and luminosity.

Required Inputs: Luminosity (erg/s) and FWHM (km/s).

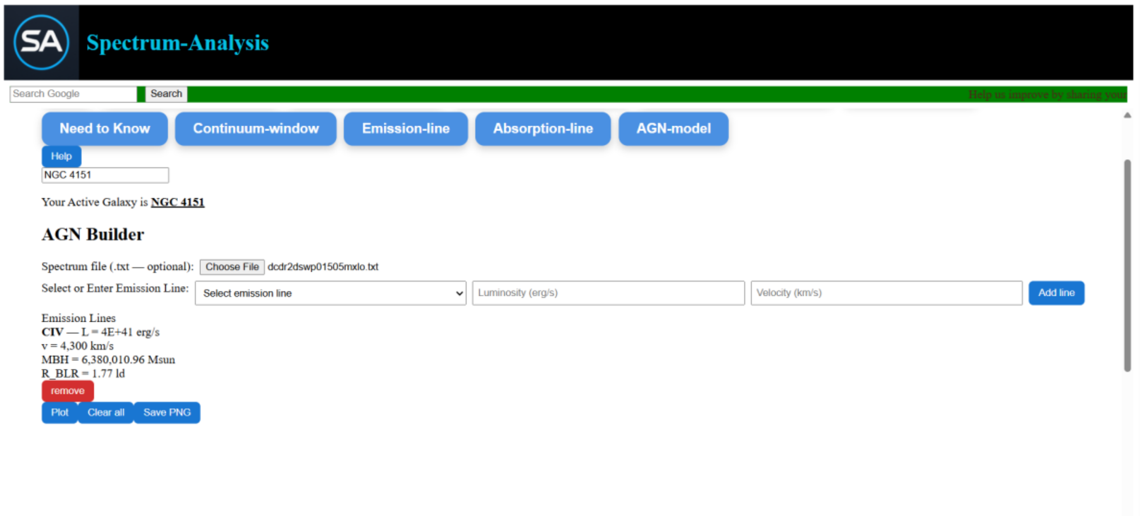
Formula Example:

- $M\_BH = f \times (R\_BLR \times FWHM^2 / G)$

8. Example Workflow

Example: Analyze the galaxy NGC 4151
1. Upload its UV spectrum (.txt file).
2. Perform continuum fitting at 1300:1350 and 2900:2950 ...
3. Fit C IV λ1549 emission line.
4. Input line luminosity and FWHM to compute M_BH and R_BLR.

Figure 7:



9. Troubleshooting & Tips

- Ensure the uploaded file is in plain text (.txt) format with two numeric columns.
- If no output appears, refresh the page and try again.
- Check browser console for any data loading errors.
- Use shorter continuum windows if the fit fails.

10. Contact & Support

For support, suggestions, or bug reports, contact:

Khundongbam Roshan Chenglei

Email: roshanchenglei@gmail.com

GitHub: https://roshanchenglei.github.io/SA

Formula Used:

1. Continuum model (power-law)
Continuum function

$$F_\lambda(\lambda) = A\,\lambda^{\alpha}$$

•A →`amplitude`
•α→ `spectral_index`

2. Continuum fitting (log–log linear regression)
Log transformation
Slope (spectral index)

$$\alpha = \frac{n\sum x_i\, y_i - \sum x_i \sum y_i}{n\,\sum x_i^2 - \left(\sum x_i\right)^2}$$

Intercept

$$\ln A = \frac{\sum y_i - \alpha \sum x_i}{n}$$

Amplitude

$$A = e^{\ln A}$$

3. Errors in continuum parameters
Variance of residuals

$$\sigma^2 = \frac{1}{n-2}\sum_{i=1}^{n}[y_i - (\alpha x_i + \ln A)]^2$$

Spectral index error

$$\sigma_\alpha = \sqrt{\frac{\sigma^2}{\sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n}}}$$

Intercept error

$$\sigma_{\ln A} = \sqrt{\sigma^2\left(\frac{1}{n} + \frac{\left(\sum x_i\right)^2}{n\left(\sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n}\right)}\right)}$$

Amplitude error

$$\sigma_A = A\,\sigma_{\ln A}$$

5. Continuum subtraction

$$F_{sub}(\lambda) = F_{obs}(\lambda) - F_{\lambda}(\lambda)$$

6. Gaussian emission line model

Single Gaussian

$$G(\lambda) = a \exp\left[-\frac{(\lambda - b)^2}{2c^2}\right]$$

- a →amplitude
- b→ central wavelength
- c→ Gaussian sigma (Å)

Total model

$$F_{model}(\lambda) = \sum_k G_k\,(\lambda)$$

7. Emission line flux

Numerical (trapezoidal) integration

$$F = \int_{\lambda_{\min}}^{\lambda_{\max}} G(\lambda)\, d\lambda \approx \sum_{j} \frac{G_j + G_{j+1}}{2} \left(\lambda_{j+1} - \lambda_j\right)$$

Analytic fallback

$$F = a\, c\sqrt{2\pi}$$

Absolute flux (IRAF style)

$$F_{abs} = | F |$$

8. Continuum RMS ($\sigma_{(C)}$)

Normalized residuals

$$r_i = \frac{F_{sub}(\lambda_i)}{\langle F_\lambda \rangle}$$

Sigma-clipped standard deviation

$$\sigma_c = std(r_i)$$

MAD fallback

$$\sigma_c = 1.48 \times median(| r_i |)$$

9. Flux error (IRAF noise formula)

$$\sigma_F = \sqrt{N}\ \sigma_c\ \Delta\lambda\ \langle F_\lambda \rangle$$

•N →number of pixels

•$\Delta\lambda$→ dispersion (Å/pixel)

10. Equivalent Width (EW)

Definition

$$\mathrm{EW} = \int_{\lambda_{\min}}^{\lambda_{\max}} \frac{F_{sub}(\lambda)}{F_\lambda(\lambda)}\, d\lambda$$

Numerical form

$$\mathrm{EW} \approx \sum_j \frac{1}{2}\left(\frac{F_{sub,j}}{F_{\lambda,j}} + \frac{F_{sub,j+1}}{F_{\lambda,j+1}}\right)\Delta\lambda$$

EW error

$$\sigma_{EW} = |\ EW\ | \frac{\sigma_F}{|\ F\ |}$$

11. FWHM
In Ångström

$$\mathrm{FWHM}_{\lambda} = 2.3548\,c$$

In velocity (km s⁻¹)

$$\mathrm{FWHM}_{v} = \frac{FWHM_{\lambda}}{b}\,c_{light}$$

Where,

$$c_{light} = 299\,792.458\ \ km\,s^{-1}$$

12. Reduced χ² (per Gaussian)

$$\chi^2 = \sum \frac{\left(F_{sub} - G\right)^2}{\sigma_{flux}^2}$$

$$\chi_{\nu}^2 = \frac{\chi^2}{N-3}$$

(3parameters: a, b, c)

13. Global reduced χ²

$$\chi_{\nu,global}^2 = \frac{1}{N_g}\sum_{k=1}^{N_g}\chi_{\nu,k}^2$$

14. Wing wavelength (velocity → Å)

Half-width

$$\Delta\lambda = \frac{\lambda_0 (FWHM/2)}{c_{light}}$$

Blue & red wings

$$\lambda_{blue} = \lambda_0 - \Delta\lambda$$
$$\lambda_{red} = \lambda_0 + \Delta\lambda$$